Climate CHANGE per interval − four ponds through the Holocene, survey plots in the Anthropocene

Holocene = UNCORRECTED CHELSA−TraCE21k (the series the analysis uses), 500−yr rolling mean across the 5 moving−window offsets, which are drawn faint behind.
Right of the dotted line: Anthropocene change across survey plots (present − past), one box per site. All paces are PER CENTURY at both ends.

Site (Anthropocene) MORA NOCA Pond (Holocene) JayBath LogWallow PantherPotholes ReflectionPond

Change over the interval (degC or mm)

MAT

degC per century

	min	mean	max
JayBath	−0.072	0.007	0.084
LogWallow	−0.073	0.007	0.084
PantherPotholes	−0.071	0.005	0.082
ReflectionPond	−0.068	0.007	0.087

MORA	1.873	2.390	2.861
NOCA	1.467	2.133	2.636

AP

mm per century

	min	mean	max
JayBath	−11.2	−1.513	9.680
LogWallow	−11.7	−1.637	10.0
PantherPotholes	−5.280	−0.323	4.000
ReflectionPond	−11.4	−1.669	9.560

MORA	−2393	−1527	−904
NOCA	−745	301	1359

MSpT

degC per century

	min	mean	max
JayBath	−0.110	0.007	0.117
LogWallow	−0.104	0.007	0.120
PantherPotholes	−0.128	0.005	0.111
ReflectionPond	−0.109	0.007	0.117

MORA	1.223	2.277	3.054
NOCA	−1.167	0.780	2.163

SpP

mm per century

	min	mean	max
JayBath	−5.000	−0.159	3.240
LogWallow	−5.040	−0.172	3.400
PantherPotholes	−2.080	0.002	1.440
ReflectionPond	−4.880	−0.165	3.320

MORA	−320	−198	−75.4
NOCA	201	480	807

MST

degC per century

	min	mean	max
JayBath	−0.091	−0.016	0.051
LogWallow	−0.092	−0.017	0.050
PantherPotholes	−0.107	−0.017	0.047
ReflectionPond	−0.103	−0.017	0.054

MORA	2.021	2.404	2.804
NOCA	1.869	2.299	2.635

SP

mm per century

	min	mean	max
JayBath	−2.440	−0.197	4.000
LogWallow	−2.520	−0.198	3.960
PantherPotholes	−1.480	−0.104	2.560
ReflectionPond	−2.400	−0.198	3.760

MORA	−257	−163	−90.0
NOCA	13.3	173	456

MWT

degC per century

	min	mean	max
JayBath	−0.133	0.014	0.160
LogWallow	−0.137	0.015	0.161
PantherPotholes	−0.138	0.011	0.144
ReflectionPond	−0.129	0.015	0.153

MORA	1.295	1.880	3.049
NOCA	0.887	1.583	2.388

WP

mm per century

	min	mean	max
JayBath	−7.040	0.190	7.360
LogWallow	−7.480	0.150	7.520
PantherPotholes	−2.600	0.306	3.200
ReflectionPond	−7.920	0.093	7.880

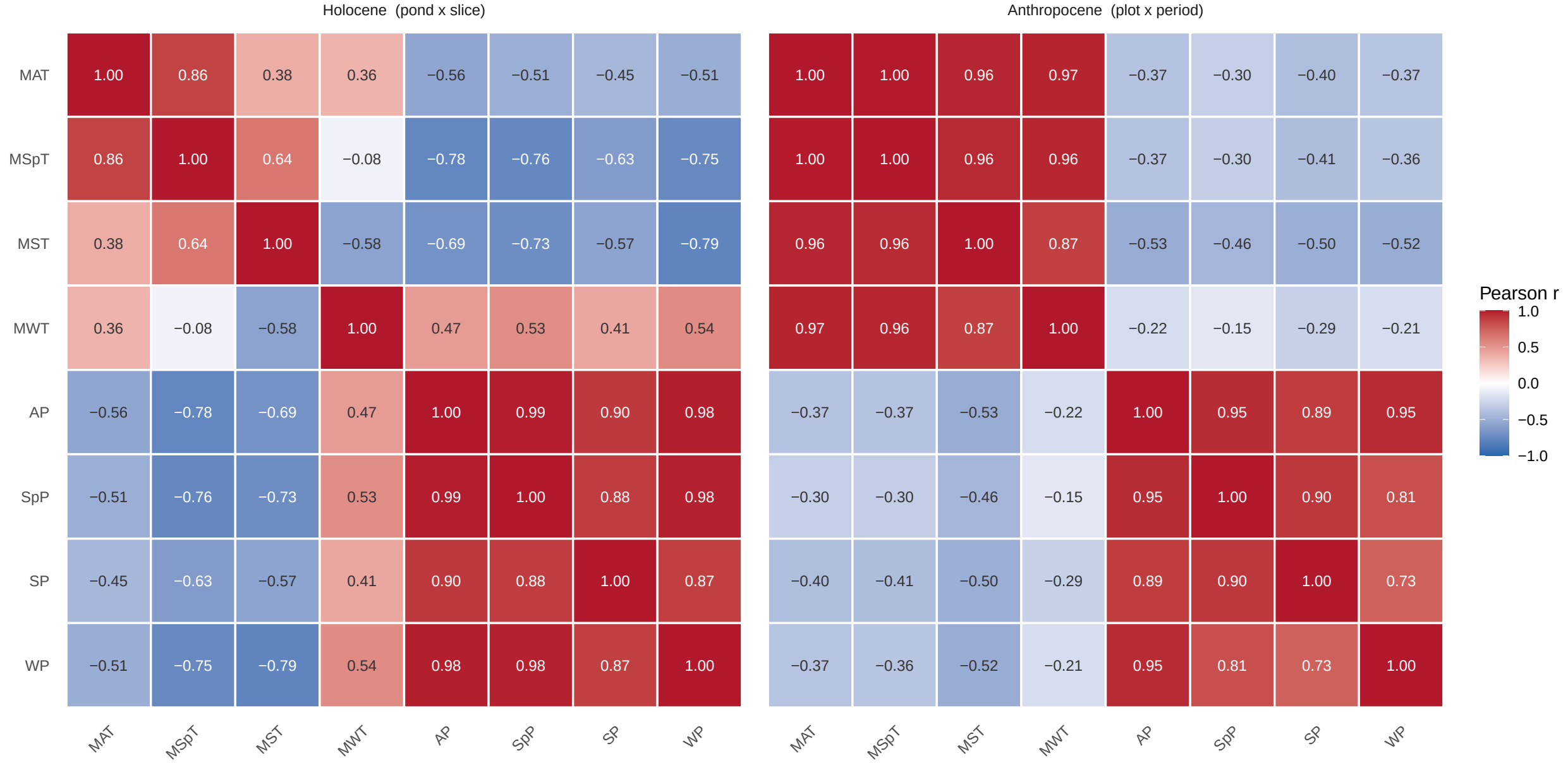
MORA	−1182	−741	−376
NOCA	−908	−415	−188

cal BP (time runs left to right) | Anthropocene

Holocene = the FULL TraCE21k record at each pond (all 111 slices, ~100 yr apart), NOT the ages where cores happen to have samples − so the line is the location's
climate history, independent of coring. Cores themselves reach JayBath 6841, LogWallow 6808, PantherPotholes 10609, ReflectionPond 5932 cal BP. Intervals differ (500 yr vs ~30 yr): these are CHANGES, not rates − see the

Correlation between climate variables, Holocene vs Anthropocene

The CCA learns the species−climate relationship from the ANTHROPOCENE plots and applies it to HOLOCENE samples.
A pair that is redundant (|r| near 1) wastes a constraint; a pair that FLIPS SIGN between the panels breaks the transfer.



Sign flips with |r| > 0.2 in both eras: MWT~MST (−0.58 vs 0.87); MST~MWT (−0.58 vs 0.87); AP~MWT (0.47 vs −0.22); SP~MWT (0.41 vs −0.29); WP~MWT (0.54 vs −0.21); MWT~AP (0.47 vs −0.22); MWT~SP (0.41 vs −0.